

Project Presentation of Co-Op Project at Industry (Module-2) (22CS421)

On

Lintellect: Code Review Agent

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- **Background and Motivation:** In the fast-paced world of modern software development, the **Pull Request (PR)** review process is often the biggest bottleneck in the CI/CD pipeline. Human reviewers are frequently overwhelmed by high-volume code changes, leading to two common failures: "rubber-stamping" PRs without deep scrutiny or focusing exclusively on superficial style nits while missing critical logical flaws.
- **Objectives:**
 - **Centralized AI-Driven Review Platform:** Develop a FastAPI-based agent that automatically detects programming languages and provides context-aware reviews for diverse codebases (Python, JS/TS, Go, Java, etc.).
 - **Multi-Stage Reasoning Workflow:** Implement a **StateGraph** architecture to decompose the review process into four distinct phases: initial analysis, issue discovery, false-positive validation, and executive reporting.
 - **Seamless Pipeline Integration:** Provide a secure, CORS-enabled API designed to trigger automatically during GitLab/GitHub pipeline runs, delivering actionable feedback.

In modern software engineering, the code review process is frequently the primary bottleneck in the delivery lifecycle. While automated linters handle syntax and formatting, they lack the contextual reasoning required to identify complex logical flaws, leaving human reviewers burdened with high-volume, repetitive tasks. **Lintellect** is an automated code review agent designed to bridge this gap. Utilizing a multi-stage **StateGraph** architecture, it simulates a senior developer's reasoning to provide deep-logic analysis directly within the CI/CD pipeline.

- **Overview: Lintellect** is an intelligent, agentic code review platform designed to act as a helper within the CI/CD pipeline. When a developer pushes code to GitLab or GitHub, Lintellect automatically intercepts the diff, detects the programming language, and executes a four-node analysis graph: **Analyzer**, **Issue Finder**, **Validator**, and **Report Generator**
- **Features:**
 - Multi-Agent Reasoning Graph
 - Automated Language Detection
 - Seamless Pipeline Integration
- **Future Enhancements:**
 - Cross-File Contextual Awareness
 - Automated "Fix-It" Suggestions

Sample Review



The screenshot displays a GitLab merge request (MR) interface. The left sidebar shows the project navigation menu with 'Merge requests' selected. The main content area shows the merge request details for 'Projects-1e19 / PRReviewAgent / Merge requests / 113'. The 'Activity' tab is active, showing a review by 'Sneha @bhardwaj@sneha111' from 1 week ago. The review includes a summary, critical issues, warnings, and recommendations.

Activity

Sneha @bhardwaj@sneha111 1 week ago

Summary

The code generates a report based on the `state` dictionary, but it contains errors. The overall quality is affected by type mismatches and potential runtime errors.

Critical Issues

- The line `issues_text = '\n'.join(state['issues'].split(','))` will raise an `AttributeError` because the `split` method is being called on a list (`state['issues']`), which does not have this method.

Warnings

- The line `issues_text = '\n'.join(state['issues'].split(','))` assumes that `state['issues']` is a string, but based on the original code, it is expected to be a list.

Recommendations

- Remove the `split` method call from the line `issues_text = '\n'.join(state['issues'].split(','))` to fix the `AttributeError` occurring in `issues_text = '\n'.join(state['issues'])`.

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Complete multi-step tasks with flows

- Review a merge request**
Identify code improvements
- Fix a failing pipeline**
Analyze pipeline failures and get fix suggestions

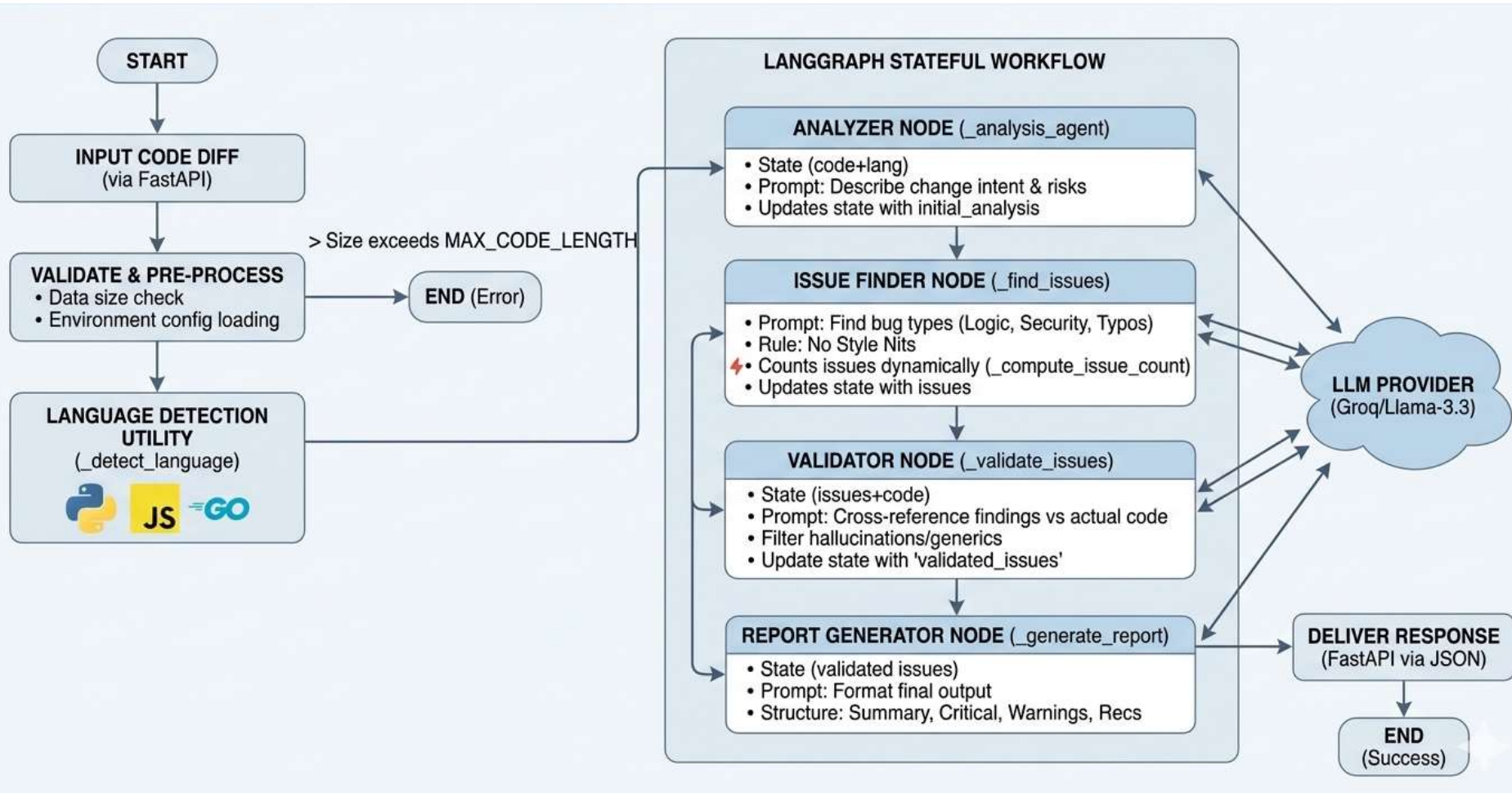
Chat with a specialized agent

With a trial you can chat with AI agents tailored to your development workflow, such as the following agents:

- Security Agent**
- Planning Agent**
- Data Analyst**

- **System Design:** The architecture follows a modular, asynchronous Service-Oriented Architecture (SOA), leveraging a stateful graph-based logic engine to process code reviews. By utilizing RESTful APIs for communication
- **Backend:**
 - FastAPI
 - LangChain
 - LangGraph

Workflow



- **Frontend:**
 - React.js
 - Tailwind CSS
- **Backend:**
 - Spring Boot
 - Docker Compose
- **Database:**
 - PostgreSQL
 - Prisma ORM
- **Supporting Tools:**
 - Docker, Render, GitHub, Postman

- Reduced Pull Request Lead Time: By automating the initial review pass, the agent eliminates the idle time where a PR sits waiting for a human reviewer.
- Enhanced Code Reliability and Security
- Unlike human reviews, which can vary based on the reviewer's mood or workload, Lintellect provides a consistent, objective baseline for every PR.

Future Prospects:

- Integration with Knowledge Graphs and RAG
- Code (Auto-Fix Capabilities) Future iterations will transition from identifying issues to proposing solutions.

References

- [FastAPI Documentation](#)
- [LangChain Documentation](#)
- [LangGraph Documentation](#)

Thank You